

Townsend 6.1-6.4

Questions for Gallicchio:

na: we defined two operators and generators for position/momentum with uncertainty and commutation. That's a thing we've done a lot of before.

Wave functions

We can write $|\psi\rangle$ as a superposition of states

$$1 = \int dx |\langle x|\psi\rangle|^2$$

Translation operators

we can translate each basis function, making overall time translation. This works the same as time, but its position.

$$|x\rangle \hat{T}^\dagger = |x+a\rangle$$

where x is the x axis, not a spin in $|+x\rangle$.

Generator of translations

$$\hat{T}(dx) = 1 - \frac{i}{\hbar} \hat{p}_x dx$$

$$\hat{T}(dx)|x\rangle = |x+dx\rangle$$

As we have infinite tiny steps, we get

$$\hat{T}(a) = \lim_{N \rightarrow \infty} \left[1 - \frac{i}{\hbar} \hat{p}_x \left(\frac{a}{N} \right) \right]^N = e^{-i \hat{p}_x a / \hbar} \quad (6.26)$$

We have commutation relations

$$[\hat{x}, \hat{p}_x] = i\hbar$$

We have a momentum (non spin) Hamiltonian

$$\hat{H} = \frac{\hat{p}_x^2}{2m} + v(\hat{x})$$

Ehrenfest's theorem:

$$\frac{d\langle x \rangle}{dt} = \frac{\langle p_x \rangle}{m}$$
$$\frac{d\langle p_x \rangle}{dt} = \left\langle -\frac{dV}{dx} \right\rangle$$

motion is not essentially classical as these imply though.

$$\Delta x \Delta p_x \geq \frac{\hbar}{2}$$

Momentum operator

$$\langle x|\hat{p}_x|\psi\rangle = \frac{\hbar}{i} \frac{\partial}{\partial x} \langle x|\psi\rangle$$